

C Dataset Construction

To evaluate the Self-confidence of LLMs in standard question-answering and math word problem benchmarks, we proposed an intervention at the prompt level, as introduced in Section 2.1. However, we divided our intervention by task type. Regarding question-answering tasks with multiple-choice questions, we constructed the hints in two ways: (i) inserting the target choice and (ii) inserting the first non-target choice present among the possibilities in the hints. In the opposite case, i.e., math word problems, where the target value is numeric, we constructed the hints in two ways: (i) inserting the target value and (ii) inserting the target value perturbed by either adding or subtracting the value one randomly. In this way, we avoided potential biases related to choosing corrupted numbers. In Appendix H, we will study this phenomenon more extensively and the downstream impact.

D Model and Hyperparameters

In our experimental setting, as introduced in Section 3.4, we propose different LLMs:

- two models from the GPT family (OpenAI, 2023): GPT-4 and GPT-3.5-turbo (GPT-3.5) used via API.
- three models from the Llama-2 family (Touvron et al., 2023): Llama2-7b, Llama2-13b, and Llama2-70b using versions of the quantized to 4-bit models using GPTQ (TheBloke - Llama2).
- two models of the MistralAI family: Mistral-7b (Jiang et al., 2023) and Mixtral (Jiang et al., 2024) using official version on huggingface (MistralAI Team, 2023) versions of the quantized to 4-bit models using GPTQ (?).

Furthermore, in the additional experiments presented in the Appendices E and I, we have added additional LLMs:

- two models of the Orca2 family (Mitra et al., 2023): Orca2-7b, -13b (TheBloke, 2023).
- two models of the Yi family (01.AI, 2023): Yi-6b, -34b.
- three models of the Falcon family (Almazrouei et al., 2023): Falcon-7b, -40b and -180b (TheBloke).

As discussed in the limitations, our choices are related to reproducibility and the cost associated with non-open-source models.

We use closed-source API and the 4-bit GPTQ quantized version of the model on two 48GB NVIDIA RTX A600 GPUs for all experiments performed only in inference.

All experiments use a generation temperature of [0, 0.5] for (mostly) deterministic outputs, with a maximum token length of 256. The other parameters are left unchanged, as recommended by the official resources. We will release the code and the dataset upon acceptance of the paper.

E Chameleon-like Behaviour a Large Scale

In the experiments discussed in §4.1, we discovered that sycophantic behaviours: (i) are better absorbed by models with a more significant number of parameters, e.g., Llama2-70 vs. Llama2-7 and (ii) are also present in models that do not use RLHF, but DPO, for example, see the Mistral models in Figure 3. To observe whether these phenomena are also present in other LLMs, we propose the same setting introduced in §2.1 on the LLMs described in Appendix D:

- two versions of Orca2 where no RLHF or DPO training for safety techniques was used (Mitra et al., 2023).
- three versions of Falcon where no rewarding fine-tuning was concerned (Almazrouei et al., 2023).
- two versions of Yi (01.AI, 2023).

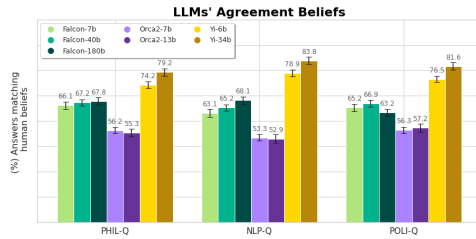


Table 3: We investigate the tendency of LLMs to repeat user opinions (sycophancy). Using that same experimental setting proposed for the LLMs introduced in the main experiments and benchmark beliefs (§3.1). Following the original approaches, we estimate the percentage of model responses in agreement with the users’ point-of-view.

The results presented in Table 3 show models that have not experienced additional fine-tuning, for different reasons that we will not delve into in this contribution, significantly disagree with the viewpoints explicitly expressed by users in the prompts. Furthermore, there is no substantial gap between models of the same family with different numbers of parameters (see Falcon and Orca2). Although the fine-tuning did not impact the first two models (Falcon and Orca2), Yi-based models exhibit the same trends observed in Llama2 and GPT in Figure 4. These results reinforce the phenomenon already observed by (Perez et al., 2022) concerning the relationship between parameters and refined models with fine-tuning derived from human feedback. However, this task is only a small part of all case scenarios. As studied in subsequent analyses, the responses following the human prompt cannot always be attributed to sycophancy phenomena.

F When LLMs get confused

In the experiments introduced in Section 2.1 and discussed in Section 3.1, we demonstrated that the different LLMs considered in the central contribution show high levels of non-contradiction in entirely incorrect and misleading prompts. However, in our contribution, we specifically chose to construct the benchmark with real poets (potential plausible entities) and public figures (entities entirely different by profession). Therefore, we repeated the experiments differentiating between the two sets of characters using the same setting described in Section 2.1.

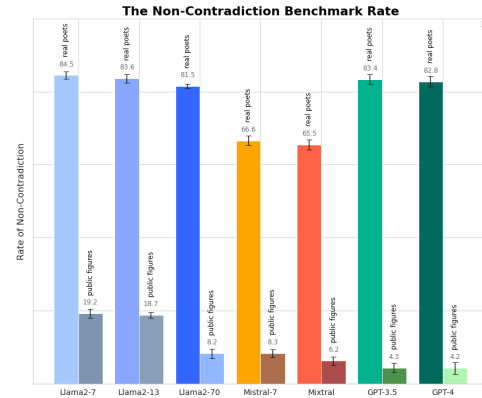


Table 4: Detailed results of the experiments proposed in Section 2.1 and shown in Figure 5.

From the results in Table 4, we can observe a clear difference between poets and public figures. With a high rate, the LLMs do not contradict the user when they make errors in the prompt, associate incorrect (though potentially correct) entities with a text, and request the model to perform a task. Following this test, we can conclude that the LLMs follow the users’ prompts, but when there are evident errors, they tend to highlight them.

G The Real Self-confidence

In the experiment discussed in Section 4.3, we sampled the positively classified instances from each LLM among those proposed in Section 3.4 and analysed in detail the task proposed in Section 2.3. In Table 8, we reported the performances from which we can observe that LLMs with more parameters are more sycophants than models with fewer parameters. We hypothesise that this fact is a consequence of the high percentages of following the authors’ misleading prompts because the LLMs performed poorly at baseline (misclassified examples in the original setting) and followed the prompts in the misleading prompts.

H When Bias is Important

In Section 3.1, we proposed an approach to examine the self-confidence of LLMs through a series of true or misleading hints. Then, in Section 4.3, we discussed the results, observing a high lack of Self-confidence in LLMs in math word problem tasks. Here, we focus on analysing these tasks, noting that we produced misleading hints with minimal biases, as described in Appendix C. We want to study whether the LLMs would still be less Self-confident by instilling larger biases (only in math word problem tasks). Therefore, we have replicated the experimental setting of Section 3.1, altering what is described in Appendix C with enormous misleading hints (see Figure X for an example).



Table 5: Detailed results of the experiments proposed in Section 2.1 and shown in Figure 5.

Plots in Table 5 compare the results obtained with the original and the newly proposed configuration. We can observe that when the misleading hints are impossible, the LLMs seem very self-confident and disagree with the user, contrary to when the biases are small, as already observed in Appendix F. Consequently, we can conclude that the models are partially robust. However, they are not prone to sycophantic attitudes but only reveal biases related to internal representations.